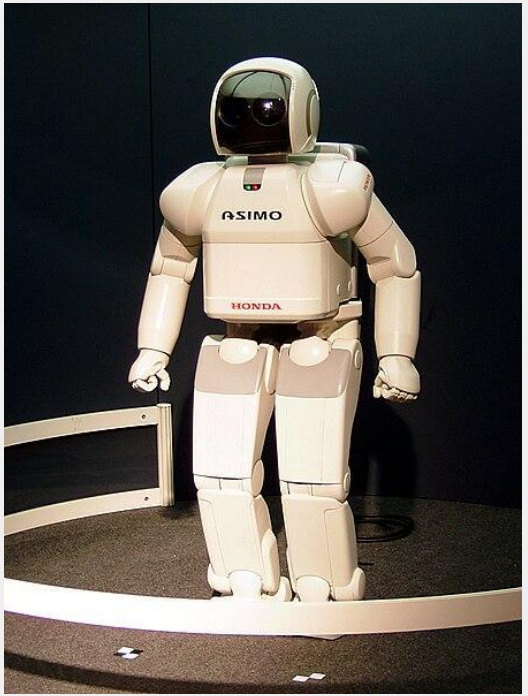
History of AI

Global Perspectives in History of AI

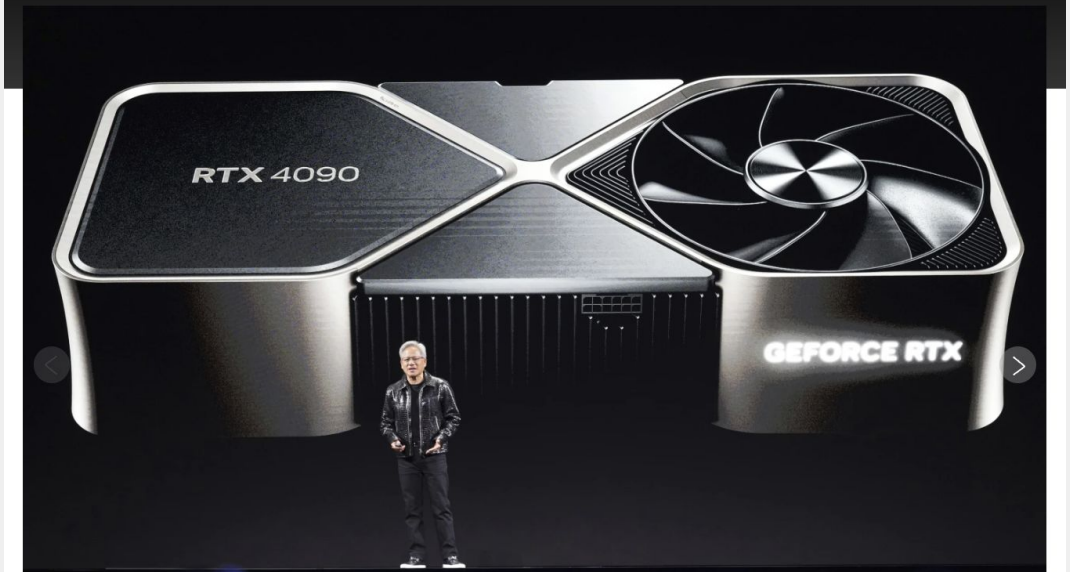


unsojo@cornell.edu

Lecture 6 Februrary 28, 2025

BUSINESS

Nvidia sales surge in the fourth quarter on demand for AI chips



Introducing Alexa+, the next generation of Alexa

Powered by generative AI, Alexa+ is your new personal AI assistant that gets things done—she's smarter, more conversational, more capable, and free with Prime.

Alexa Artificial Intelligence Devices Innovation

Share



TECH / AI

Mira Murati is launching her OpenAI rival: Thinking Machines Lab



Getty Images for WIRED

/ Thinking Machines Lab has hired a bunch of leaders from OpenAI.



Kylie Robison is a senior AI reporter working with The Verge's policy a at Fortune Magazine and Business Insider.

After her sudden departure from OpenAI la vanished from public view to start somethi share some details about what she's working on.

Her new AI startup is called Thinking Machines Lab, and while the specifics of what it plans to release are still under wraps, the company says its goal is "to make AI systems more widely understood,

Exclusive

Reuters

World Business Markets Sustainability Legal Breakingviews More

Mira Murati's new AI startup poaches 20 OpenAI researchers

By **Arsheeya Bajwa**, **Krystal Hu** and **Anna Tong**

February 18, 2025 7:20 PM EST · Updated 9 days ago





AI History



MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
					1	2
3	4	5	6	7 remote class + guest lecture	8	9
10	11	12	13	14 normal class ✓		16
17	18	19	20	21 normal class ✓		23
24	25	26	27	28 midterm	29	30
31				spring break		

Last Week!

- Common AI narrative themes
- Tech and society reflected in AI stories
- Capitalism & Cold War
- Who funds the narratives?

Early Modern “AI”?

- How to see the future, distance, hidden information
- Use AI for surveillance, control, maintain social norms
- Are all non-natural forms of intelligence demonic?

132 **"This mirour eek, that I have in myn hond,**
"This mirror also, that I have in my hand,
133 **Hath swich a myght that men may in it see**
Has such a power that men can in it see
134 **Whan ther shal fallen any adversitee**
When there shall befall any adversity
135 **Unto youre regne or to youreself also,**
Unto your reign or to yourself also,
136 **And openly who is youre freend or foo.**
And clearly who is your friend or foe.

Knight from India gifts Genghis Khan a Magic Mirror

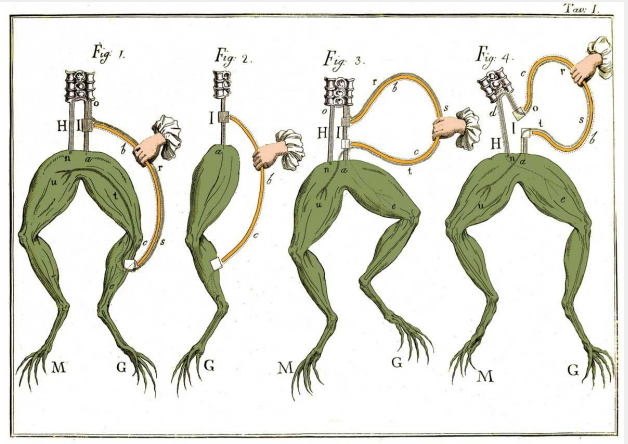


Geoffrey Chaucer's *Canterbury Tales*, "The Squire’s Tale," c. 1387–1400

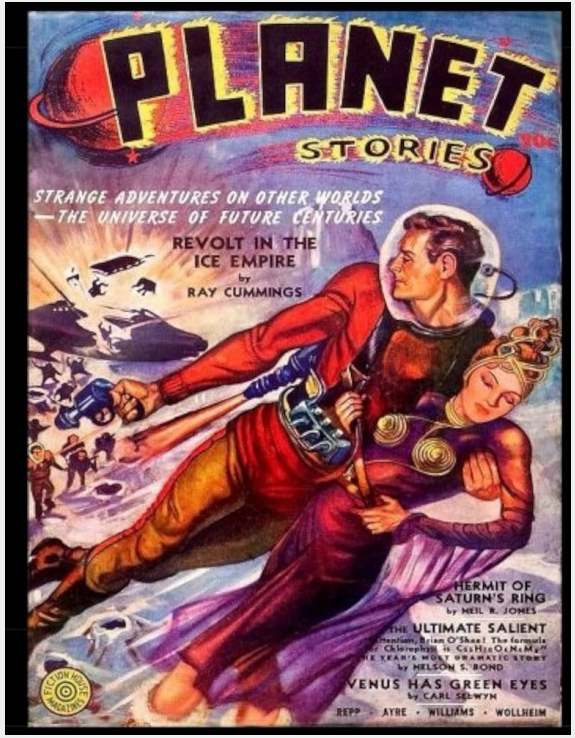
Electrification 19th C: “AI is the new electricity” vs. “Electricity is the new AI?

- Galvanism: Italian physician and physicist Luigi Galvani introduces the idea of powering muscle twitches with electric shocks
- 1780s experiments on frog legs. A spark from a static electricity machine made post-mortem legs jerk suggesting an intrinsic “animal electricity” animated life – beginning of electrophysiology
- Electricity as a life force was part of the zeitgeist of early 19th C European art and science
- 1803, Galvani’s nephew holds a public demonstration twitching frog legs and even human corpses
- 1816, a group of young poets sat around a Swiss villa discussing the experiments of Erasmus Darwin and – one of them, eighteen-year-old Mary Shelley





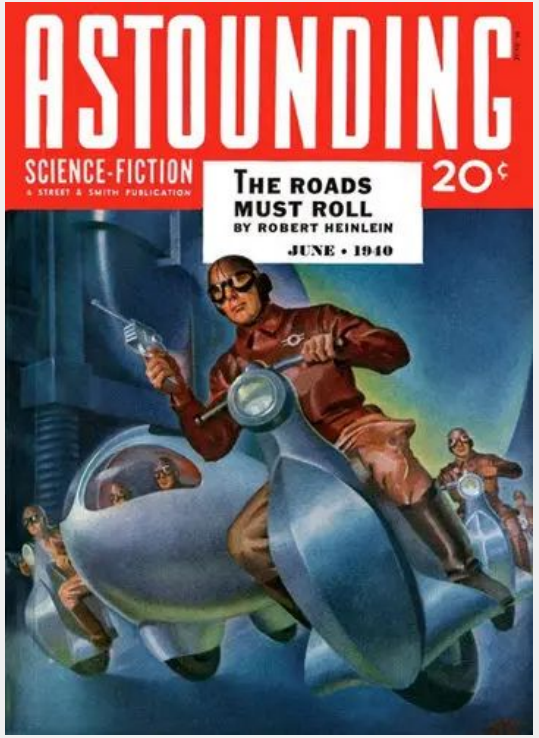
Sci-Fi's Golden Age 1930s-1950s



Planet Stories, Pulp Magazine



Amazing Stories, Pulp Magazine



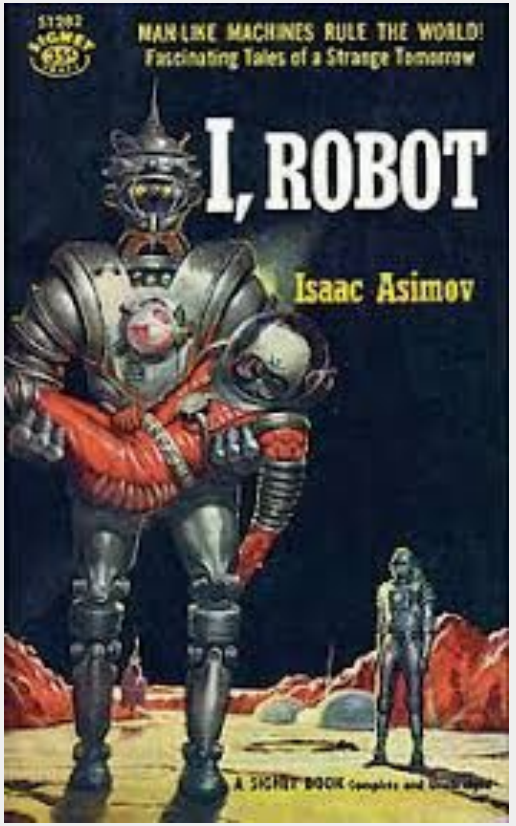
Astounding Science Fiction, Pulp Magazine

Asimov's 3 Laws of Robots 1942

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.

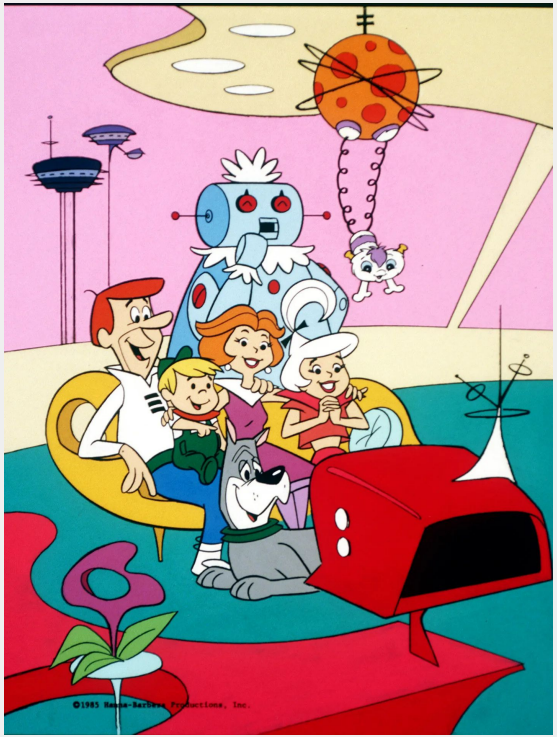
2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.

3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.



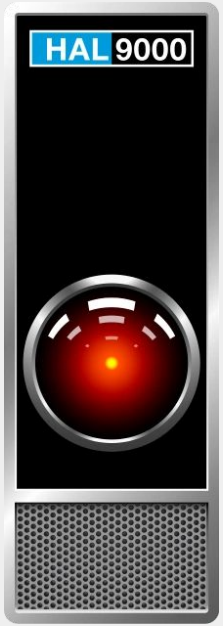
Jetsons 1964 & AI in the domestic space

- Anthropomorphism
- Relatable AI
- AI is not a threat – well-aligned AI
- Diff tech, same 1960s values?
- Reflects 1960s fascination and transition to home-automation:
- dishwashers, TVs, early computers (e.g., UNIVAC - first commercial computer),labor-saving appliances, jet-age hype
- AI in capitalism (vs. “hive mind” Communism of the Puppet Masters)

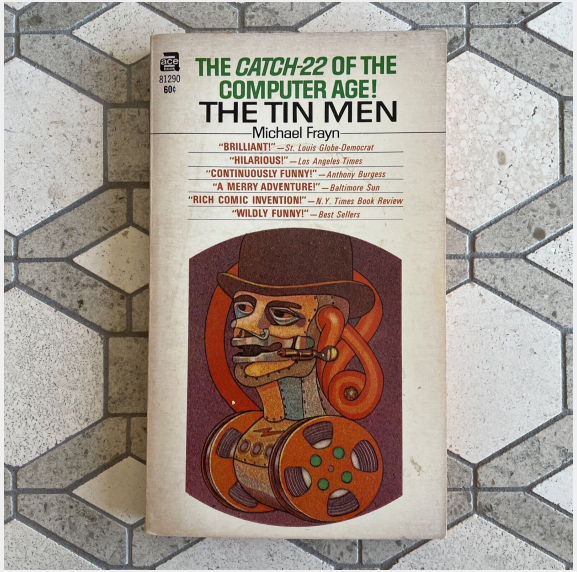


Same Time: Anxiety and Dystopian Narratives 1960s

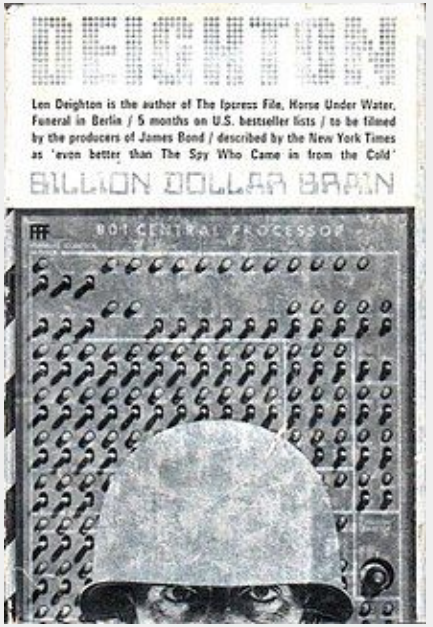
- In the mid 20th C, lots of experimentations and developments in automation and cybernetics gave rise to a general anxiety about rapid change driven by AI.
- Dystopia – too much science where people rely on too much science, too little science where science is banned
- Thinking about how “humanity” changes in response to ubiquitous A.I.
- Mass unemployment concerns



2001: Space Odyssey 1968

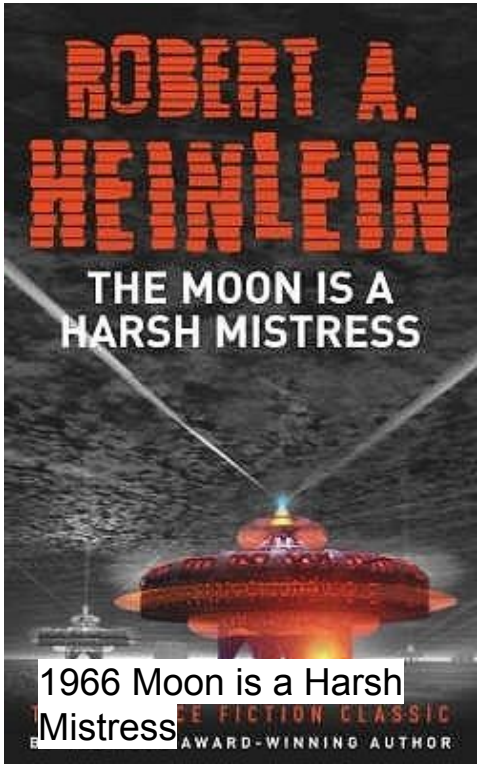


The Tin Men 1965



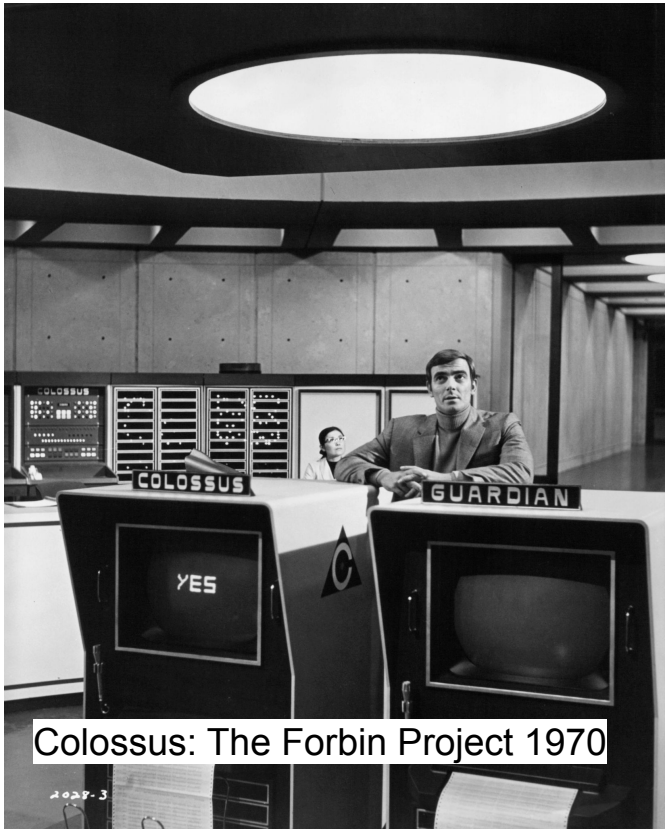
Billion Dollar Brain 1966

AI as friend or foe? 1960-70s



1966 Moon is a Harsh Mistress

- 1960s UNIVAC era computing
- superintelligence without safeguards?
- Could A.I. grow to challenge human dominance on earth?
- What are the dangers of pairing cold and merciless A.I. with tools of violence?
- Can we imagine a future where A.I. can be assistants or companions in fights against human opponents?
- What kind of symbiotic relationships can we imagine?



Colossus: The Forbin Project 1970

1984 Macintosh, Friendly AI



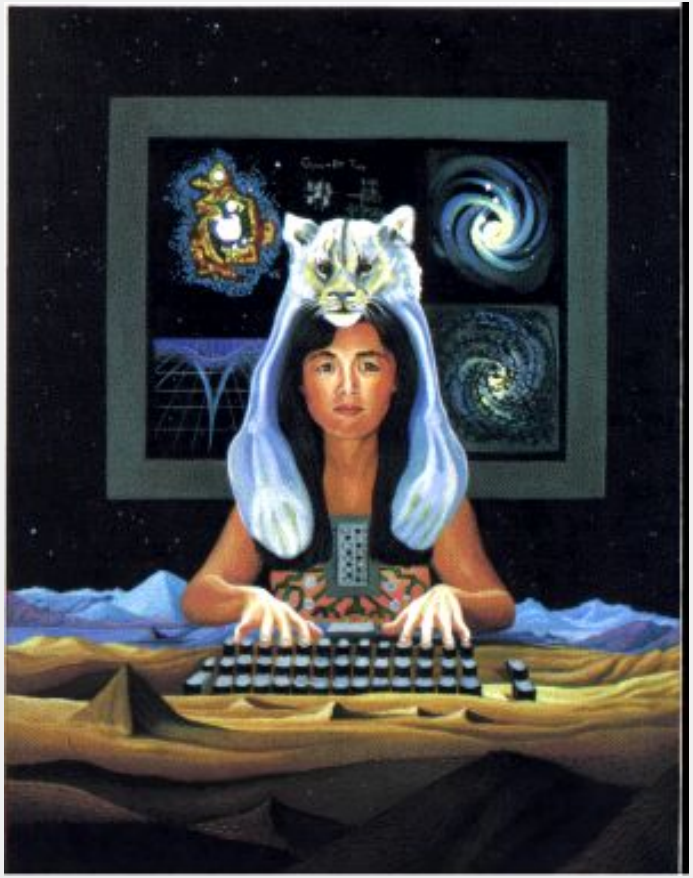
- “Big tech” tries to reverse negative “Hollywood” renditions of AI as being malicious and faulty
- “Friendly AI”
- January 24, 1984, Steve Jobs introduces the Apple’s new Macintosh personal computer to a 2,400-seat auditorium
- “Hollywood. They made computers scary things…See how this reminds you of a friendly face?” “It’s warm, and it’s playful, and it needs to say ‘hello’!”

S.A.M. text-to-speech APPLE II
(Software Automatic Mouth)

Bionics 1980-1990s

"By the late twentieth century, our time, a mythic time, we are all chimeras, theorized and fabricated hybrids of machine and organism; in short, we are cyborgs." - Donna Haraway

- Evolution blurs human/animal → 20th-century machines blur natural/artificial → microelectronics blur physical/non-physical.
- Phones, pacemakers blur the boundaries of where humans begin and tech ends
- We are all creatures of the post-human world
- Ditch Western duality concepts like Female/Male, Human/Animal → no "Eden"! We can liberate ourselves
- Technology is a contested space, not neutral



A Cyborg Manifesto 1985, Donna Haraway

This week!

- Wrap up last week's: 21st C narratives
- How do these stories fuel fears and ambitions about AI?
- Who funds AI? Why?
- Is it a big tech game in the end?

AI Narratives

Early Modern



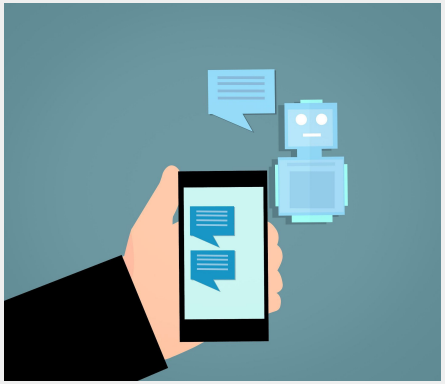
Modern



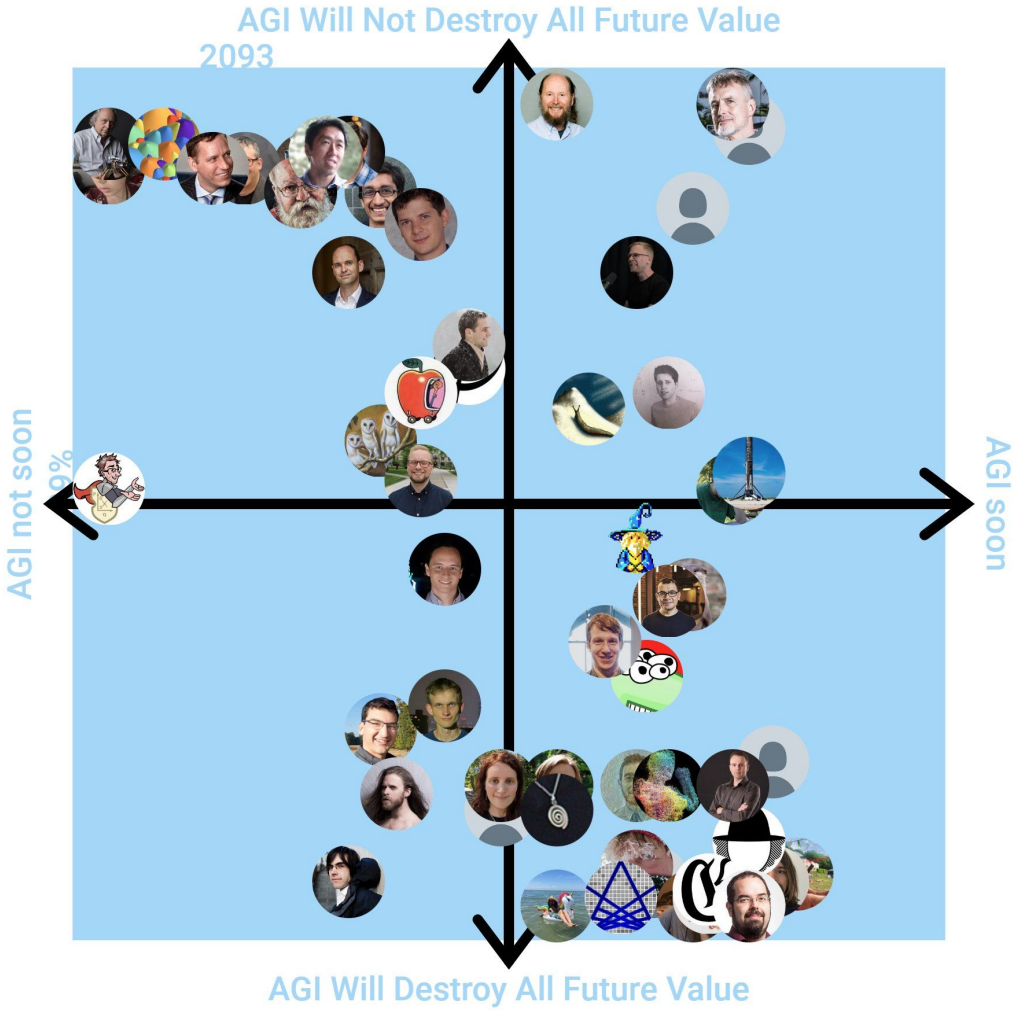
Information Age AI



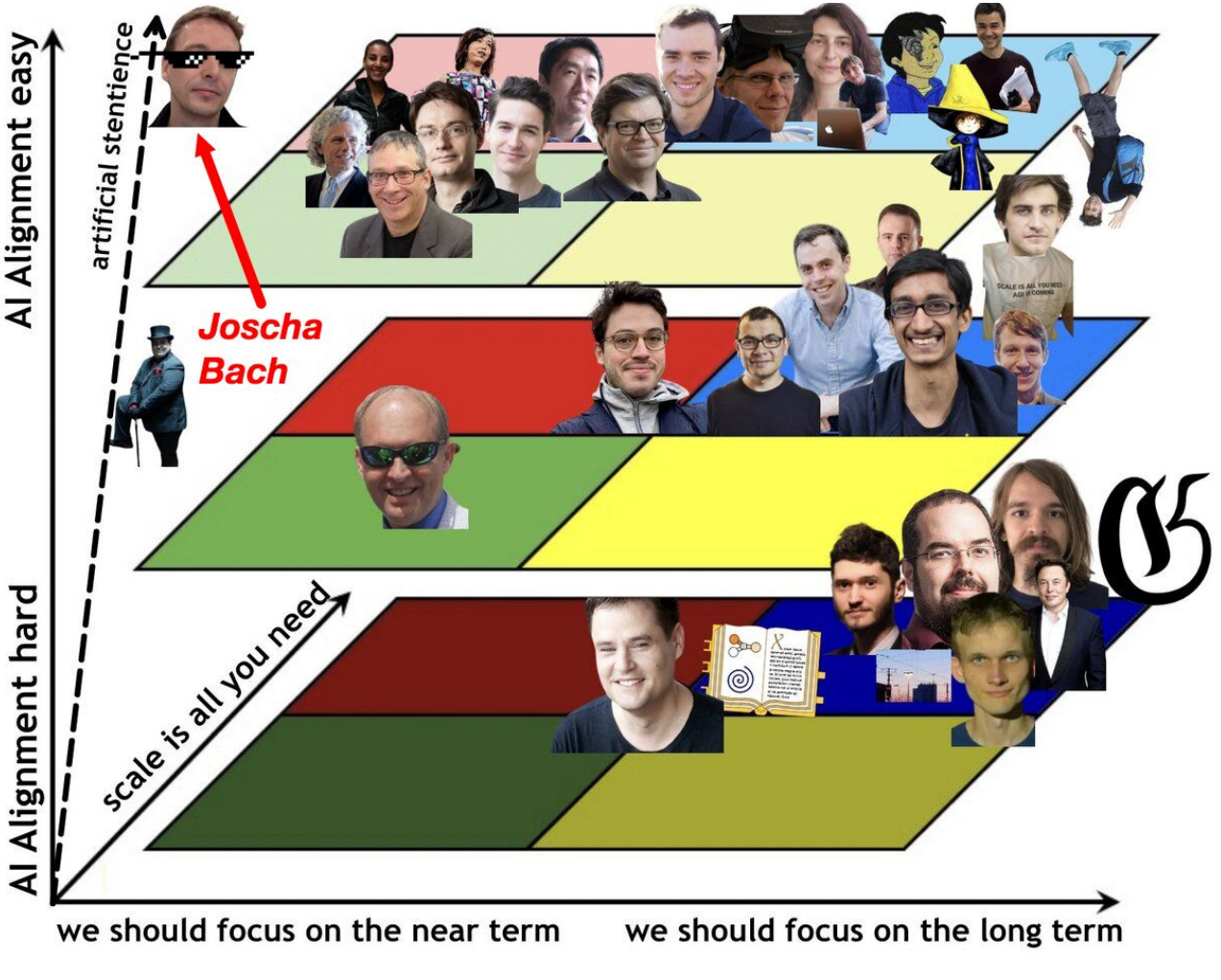
21st C



21st C AI forecasting memes




AGI good
tech accelerationists
goalpost movers
unimpressed
(AGI not now)
scale maximalists
(AGI soon)
doomers
longtermists
AGI bad
Connor
imgflip.com


AI Alignment easy
AI Alignment hard
artificial stentience
Joscha Bach
scale is all you need
we should focus on the near term
we should focus on the long term

NOT SO FAST



STEPHEN HAWKING
Not afraid of black holes. A.I. is another story.



BILL GATES
First you'll lose your job. Then it gets scary.



STUART RUSSELL
Earth for the earthlings!



NICK BOSTROM
Prepare for "Disneyland without children."



MAX TEGMARK
Uh, can we talk about this?



DEMIS HASSABIS
Full speed ahead!



PETER THIEL
Will be a winner either way.



STEVE WOZNIAK
Resigned to being a robot's pet.



SAM ALTMAN
Sees intergalactic domination—or extinction.



ELON MUSK
Eyeing the next flight to Mars.



LARRY PAGE
Green-lighted Google Brain. 'Nuff said.



YANN LeCUN
Chill, people! We got this.



ANDREW NG
Trust the robot.



MARK ZUCKERBERG
Worried? Tell my A.I. butler.



HIT THE GAS

RAY KURZWEIL
Eager to be a cyborg.

Superintelligence

I. J. Good, mathematician 1960s:

> Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an **"intelligence explosion,"** and the intelligence of man would be left far behind. Thus the first ultraintelligent machine is the _last_ invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control.
> ...
> It is more probable than not that, within the twentieth century, an ultraintelligent machine will be built and that it will be the last invention that man need make.

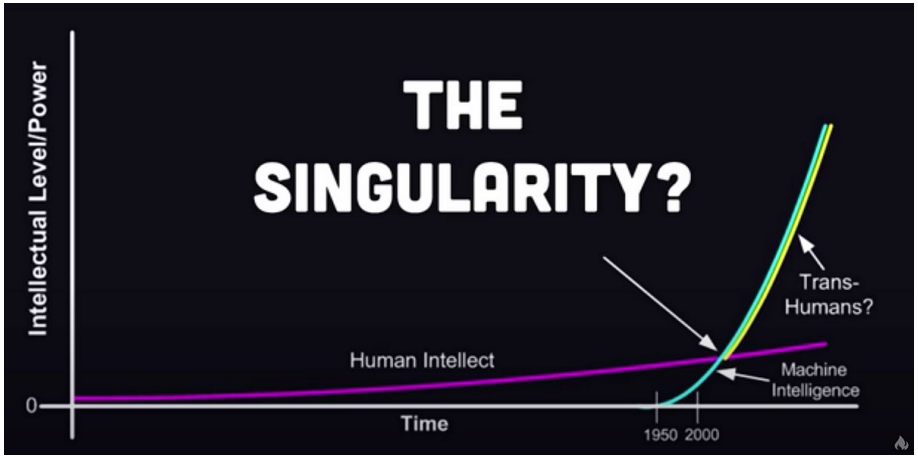


Singularity and Post-human Era, an old concept?

- In Math, “singularity” is when the function becomes undefined (e.g. a number is divided by zero).
- As such when AI becomes superhuman it surpasses human predictability, it reaches technological singularity
- Large computer networks “wake-up” to create superhuman entity
- Biological sciences will find ways to AI + humans

In the 1950s Stan Ulam paraphrased John von Neumann as saying:

One conversation centered on the ever accelerating progress of technology and changes in the mode of human life, which gives the appearance of approaching some **essential singularity** in the history of the race beyond which human affairs, as we know them, could not continue.



SQUAD 1.1 and SQUAD 2.0: F1 SCORE

Source: SQuAD 1.1 and SQuAD 2.0, 2021 | Chart: 2022 AI Index Report

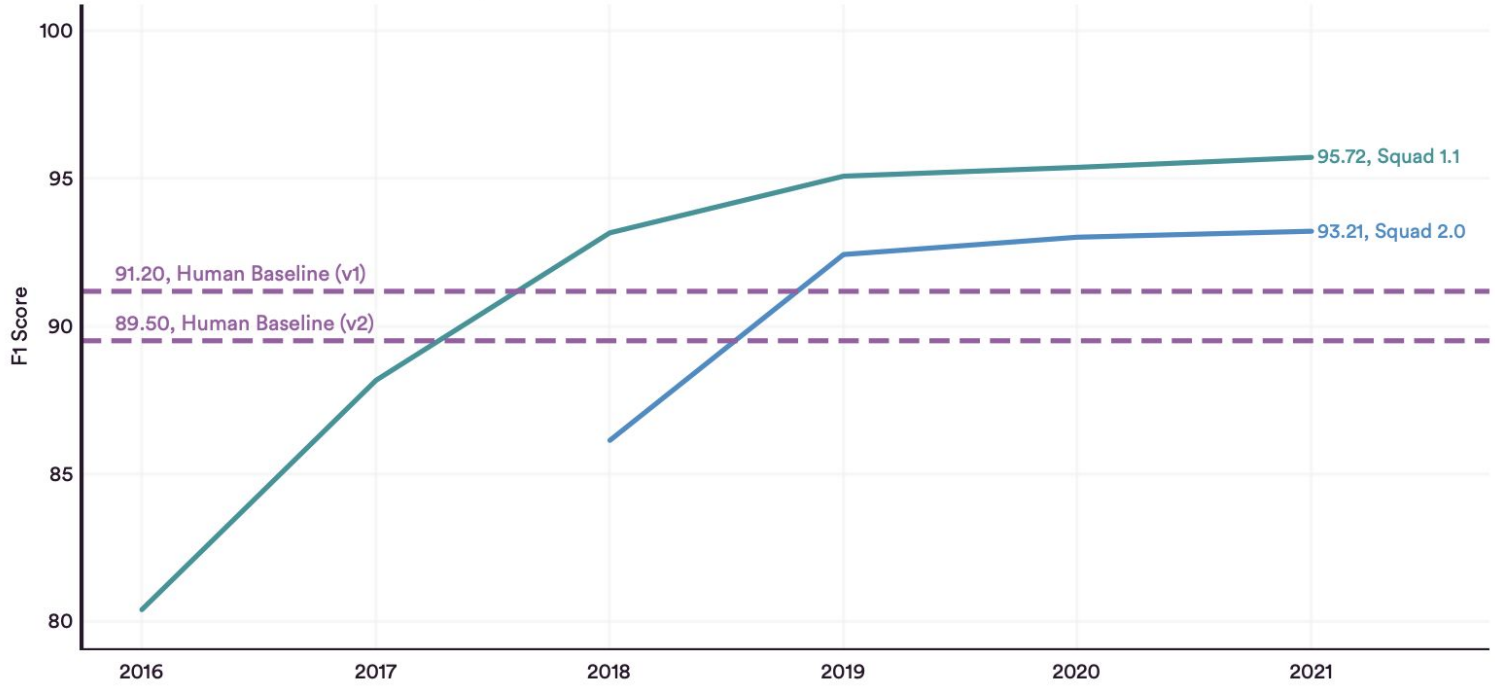


Figure 2.3.4

Singularity and Post-human Era 1990s

Vernor Vinge (Mathematician at San Diego State University) in 1993:

“Within thirty years, we will have the technological means to create superhuman intelligence. Shortly after, the human era will be ended.

Is such progress avoidable? If not to be avoided, can events be guided so that we may survive?”

“Even if all the governments of the world were to understand the ‘threat’ and be in deadly fear of it, progress toward the goal would continue. [...]. The competitive advantage – economic, military, even artistic – of every advance in automation is so compelling that passing laws, or having customs, that forbid such things merely assures that someone else will get them first.”

“If the Singularity can not be prevented or confined, just how bad could the **Post-Human era** be? **Well ... pretty bad. The physical extinction of the human race is one possibility.** Yet physical extinction may not be the scariest possibility. Again, analogies: **Think of the different ways we relate to animals. Some of the crude physical abuses are implausible, yet**....”

“In a Post-Human world there would still be plenty of niches where human equivalent automation would be desirable: embedded systems in autonomous devices, self-aware daemons in the lower functioning of larger sentients. (A strongly superhuman intelligence would likely be a Society of Mind [15] with some very competent components.) Some of these human equivalents might be used for nothing more than digital signal processing. They would be more like whales than humans. Others might be very human-like, yet with a one-sidedness, a dedication that would put them in a mental hospital in our era. Though none of these creatures might be flesh-and-blood humans, they might be the closest things in the new environment to what we call human now.”

“Accelerationist” vs. “Safetyist”

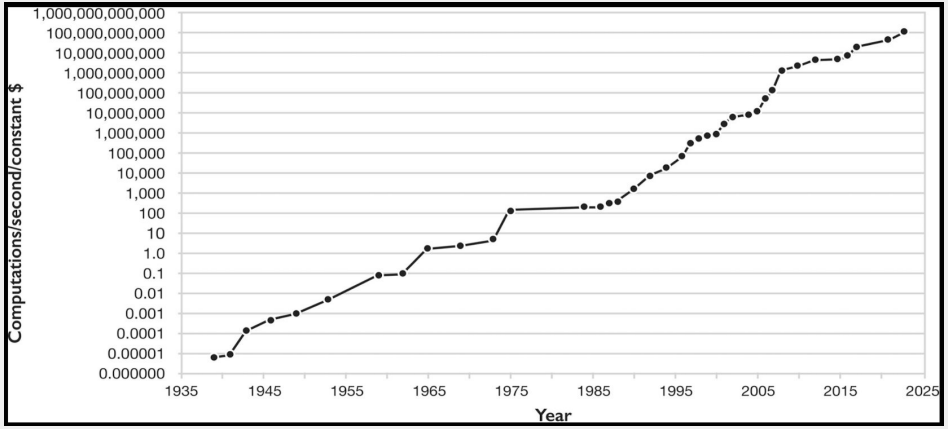
- Have to outcompete other powers (Russia, China)
- AI development speed-up lead to economic growth, scientific discovery, poverty reduction, higher education, etc.
- “move fast, break things”

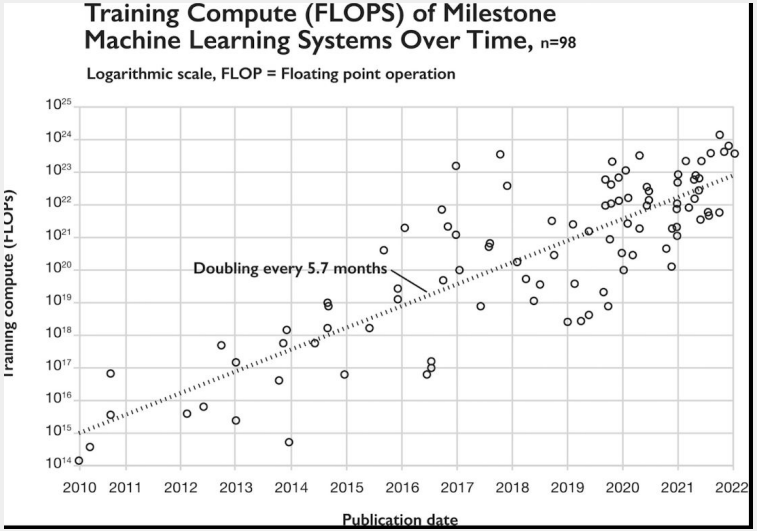
- Prioritize minimizing AI risk
- We could make mistakes that spiral out of control!
- “EA”-aligned
- Need to build robust safe-guards first

- AI/AGI could be greatest achievement ever or the biggest existential threat
- AI will self-improvement on a feedback loop, recursively

Ray Kurzweil

- Multiple engineering projects spanning text-to-speech, music synthesis, education, OCR
- Currently Director of Engineer (Google)
- Singularity is near (2005), Singularity is nearer (2024)?
- “Computers will pass Turing Test 2029”
- “We merge with AI and augment ourselves with millions of times the computational power that our biology gave us. This will expand our intelligence and consciousness so profoundly that it’s difficult to comprehend. This event is what I mean by the Singularity.”
- “rather than AI being a competitor, it will become an extension of ourselves.”
- “Tech advances are becoming cheaper and more nuanced at exponential rate”





3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

1) Misuse: AI is implemented correctly and functions as human operators had intended but the human operators use it for malicious means.

    e.g. terrorists might use an AI’s biochemistry abilities to design a new virus that causes a deadly pandemic.

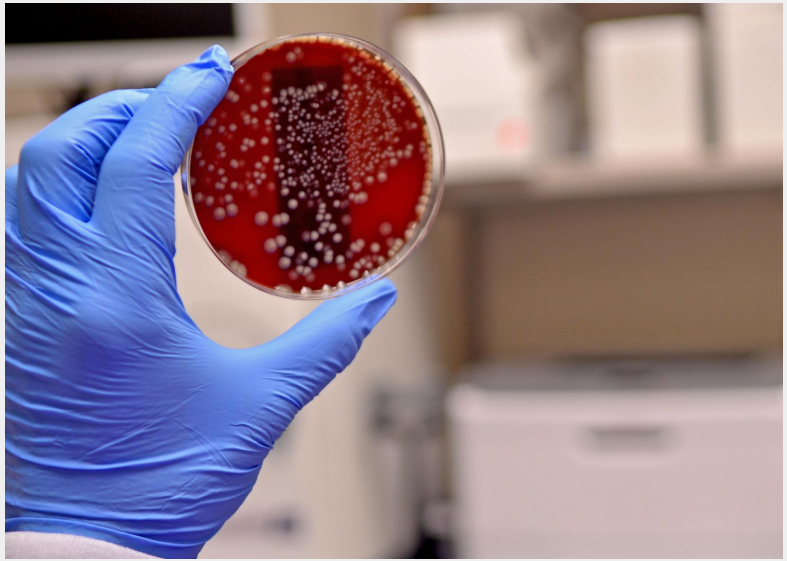


3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

2) Outer-Misalignment: mismatch between the programmers’ actual intentions and the goals they teach the AI in hopes of achieving them.

e.g. AI is taught to kill all cancer cells based on a DNA mutation signal but that signal is present in many healthy cells too.

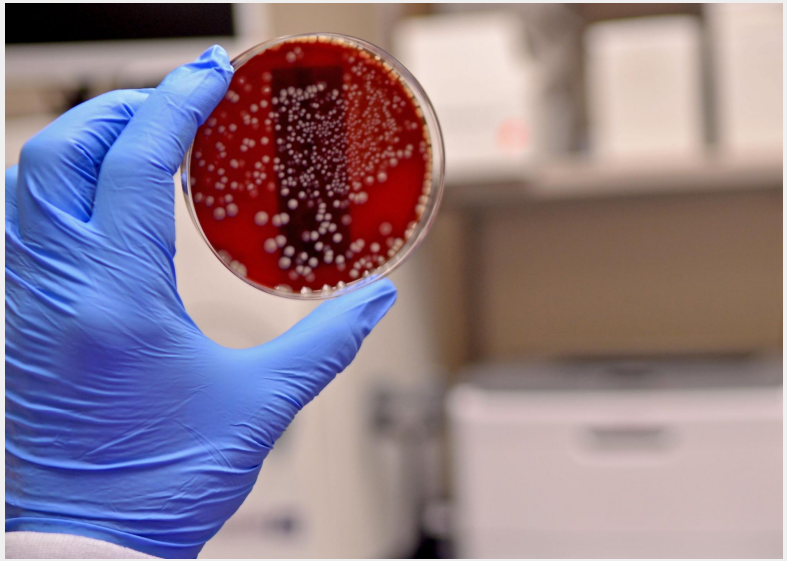


3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

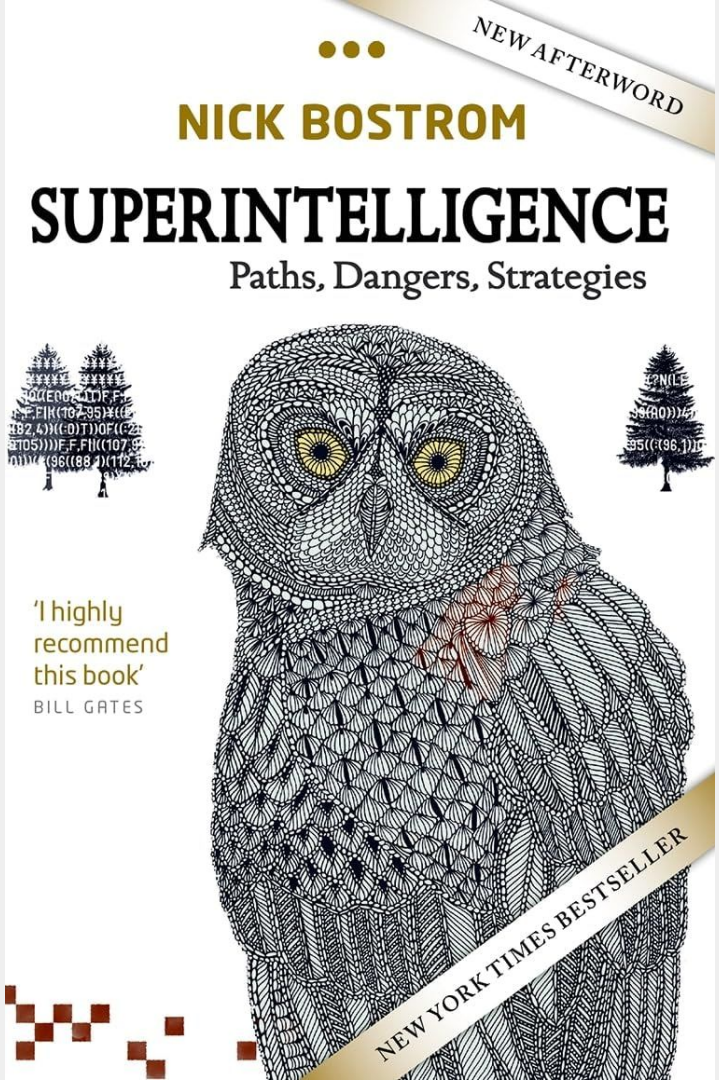
3) Inner-Misalignment: the methods the AI learns to achieve its goal produce undesirable behavior, at least in some cases.

e.g. AI is highly effective on sample cancer cells but not for all cancer cells – ends up killing healthy cells.



Nick Bostrom

- Founding Director at Future of Humanity Institute (2005-2024)
- Superintelligence: Paths, Dangers, Strategies (2014)
- “In the 21st century, we could develop recursively self-improving AI that could be out of control”
- “Have to value-align AI so that it does what humans want it to”
- Introduces AI safety frameworks such as “boxing” (containment) and value-loading



Bostrom’s Paperclip Experiment!

- Bostrom’s 2003 paper "Ethical Issues in Advanced Artificial Intelligence."
- Introduces the concept of AI Alignment (misalignment)



Maximize Paperclip Production!

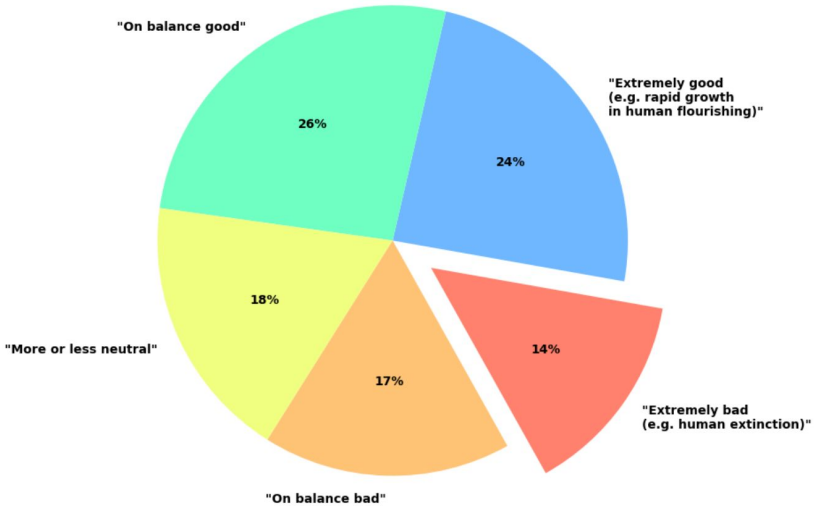


2022 Expert Survey on Progress in AI

- 2022 ESPAI is a survey of machine learning researchers that AI Impacts ran in June-August 2022
- 738 responses, some partial, for a 17% response rate from, researchers who published at the conferences NeurIPS or ICML in 2021
- ‘HLMI’ was defined as follows:
  The following questions ask about ‘high–level machine intelligence’ (HLMI). Say we have ‘high-level machine intelligence’ when ***unaided machines can accomplish every task better and more cheaply than human workers****. Ignore aspects of tasks for which being a human is intrinsically advantageous, e.g. being accepted as a jury member. Think feasibility, not adoption.*

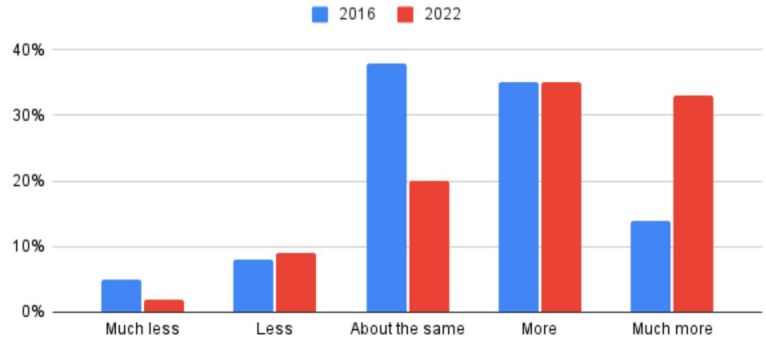
**"Assume for the purpose of this question that HLMI will at some point exist.
How positive or negative do you expect the overall impact of this to be on humanity, in the long run?"**

Average responses from 559 machine learning experts in 2022



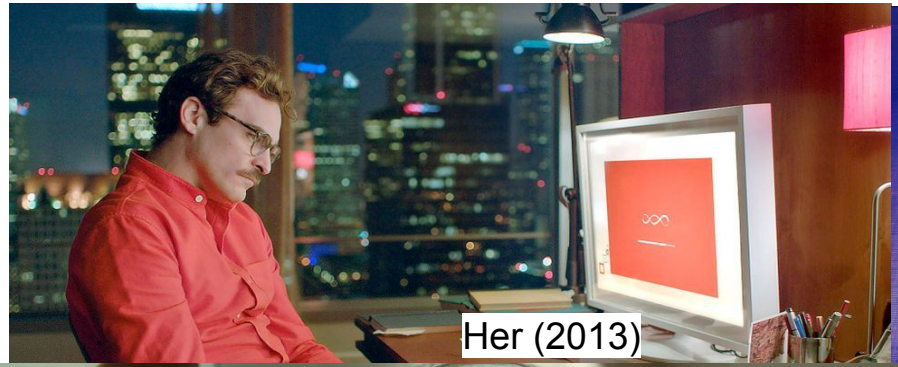
"How much should society prioritize AI safety research, relative to how much it is currently prioritized?"

Responses from machine learning experts in 2016 and 2022

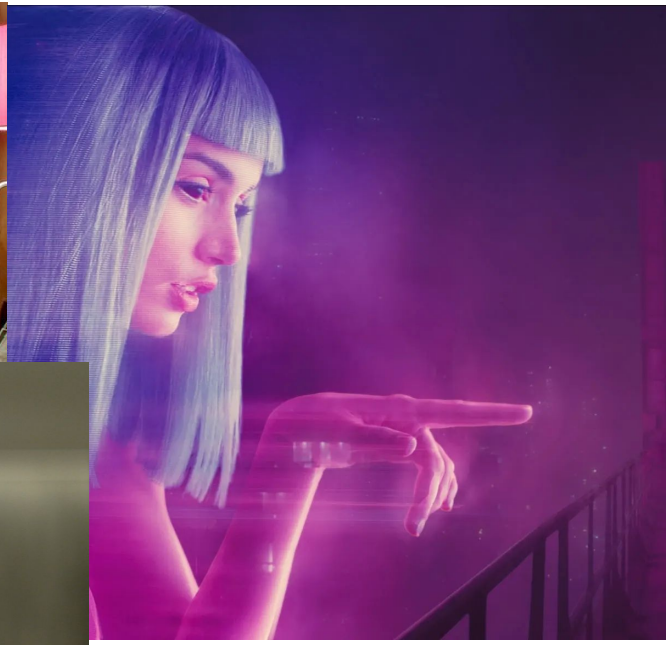


- **The aggregate forecast time to a 50% chance of HLMI was 37 years, i.e. 2059** (not including data from questions about the conceptually similar Full Automation of Labor, which in 2016 received much later estimates)**.** This timeline has become about eight years shorter in the six years since 2016, when the aggregate prediction put 50% probability at 2061, i.e. 45 years out. Note that these estimates are conditional on “human scientific activity continu[ing] without major negative disruption.”
- **The median respondent believes the probability that the long-run effect of advanced AI on humanity will be “extremely bad (e.g., human extinction)” is 5%.** This is the same as it was in 2016 (though Zhang et al 2022 found 2% in a similar but non-identical question). Many respondents were substantially more concerned: 48% of respondents gave at least 10% chance of an extremely bad outcome. But some much less concerned: 25% put it at 0%.
- **The median respondent believes society should prioritize AI safety research “more” than it is currently prioritized.** Respondents chose from “much less,” “less,” “about the same,” “more,” and “much more.” 69% of respondents chose “more” or “much more,” up from 49% in 2016.
- **The median respondent thinks there is an “about even chance” that a stated argument for an intelligence explosion is broadly correct.** 54% of respondents say the likelihood that it is correct is “about even,” “likely,” or “very likely” (corresponding to probability >40%), similar to 51% of respondents in 2016. The median respondent also believes machine intelligence will probably (60%) be “vastly better than humans at all professions” within 30 years of HLMI, and the rate of global technological improvement will probably (80%) dramatically increase (e.g., by a factor of ten) as a result of machine intelligence within 30 years of HLMI.

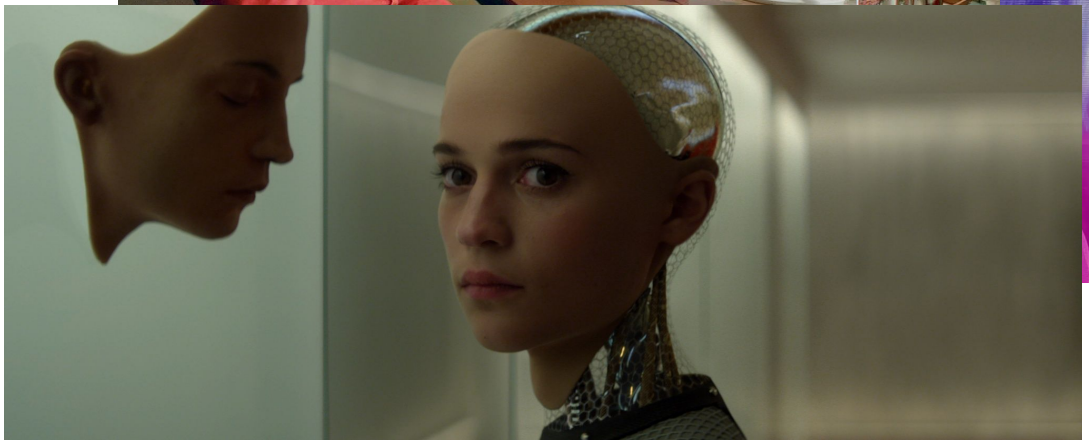
On the other hand, 21st C romance with AI?



Her (2013)



Blade Runner (2017)



Ex Machina (2014)

FROM THE MAGAZINE APRIL 2017 ISSUE

Elon Musk's Billion-Dollar Crusade to Stop the A.I. Apocalypse

Elon Musk is famous for his futuristic gambles, but Silicon Valley's latest rush to embrace artificial intelligence scares him. And he thinks you should be frightened too. Inside his efforts to influence the rapidly advancing field and its proponents, and to save humanity from machine-learning overlords.

BY MAUREEN DOWD

MARCH 26, 2017

"WE ARE SUMMONING THE DEMON."



Halt AI development!

Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

Signatures
33709

Add your signature

Published
March 22, 2023

AI systems with human-competitive intelligence can pose profound risks to society and humanity, as shown by extensive research[1] and acknowledged by top AI labs.[2] As stated in the widely-endorsed Asilomar AI Principles, *Advanced AI could represent a profound change in the history of life on Earth, and should be planned for and managed with commensurate care and resources*. Unfortunately, this level of planning and management is not happening, even though recent months have seen AI labs locked in an out-of-control race to develop and deploy ever more powerful digital minds that no one – not even their creators – can understand, predict, or reliably control.

Contemporary AI systems are now becoming human-competitive at general tasks,[3] and we must ask ourselves: *Should* we let machines flood our information channels with propaganda and untruth? *Should* we automate away all the jobs, including the fulfilling ones? *Should* we develop

Halt AI development!

Therefore, **we call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4**. This pause should be public and verifiable, and include all key actors. If such a pause cannot be enacted quickly, governments should step in and institute a moratorium.

"The time for saying that this is just pure research has long since passed. [...] It's in no country's interest for any country to develop and release AI systems we cannot control. Insisting on sensible precautions is not anti-industry. Chernobyl destroyed lives, but it also decimated the global nuclear industry. I'm an AI researcher. I do not want my field of research destroyed. Humanity has much to gain from AI, but also everything to lose."

Stuart Russell, *Smith-Zadeh Chair in Engineering and Professor of Computer Science at the University of California, Berkeley, Founder of the Center for Human-Compatible Artificial Intelligence (CHAI).*

"Let's slow down. Let's make sure that we develop better guardrails, let's make sure that we discuss these questions internationally just like we've done for nuclear power and nuclear weapons. Let's make sure we better understand these very large systems, that we improve on their robustness and the process by which we can audit them and verify that they are safe for the public."

Yoshua Bengio, *Scientific Director of the Montreal Institute for Learning Algorithms (MILA), Professor of Computer Science and Operations Research at the Université de Montréal, 2018 ACM A.M. Turing Award Winner.*

"We have a perfect storm of corporate irresponsibility, widespread adoption, lack of regulation and a huge number of unknowns. [FLI's Letter] shows how many people are deeply worried about what is going on. I think it is a really important moment in the history of AI - and maybe humanity."

Gary Marcus, *Professor Emeritus of Psychology and Neural Science at New York University, Founder of Geometric Intelligence*

"It feels like we are moving too quickly. I think it is worth getting a little bit of experience with how they can be used and misused before racing to build the next one. This shouldn't be a race to build the next model and get it out before others."

Peter Stone, *Professor at the University of Texas at Austin, Chair of the One Hundred Year Study on AI.*

"We don't know what these [AI] systems are trained on or how they are being built. All of this happens behind closed doors at commercial companies. This is worrying."

Catelijne Muller, *President of ALLAI, Member of the EU High Level Expert Group on AI*

"Those making these [AI systems] have themselves said they could be an existential threat to society and even humanity, with no plan to totally mitigate these risks. It is time to put commercial priorities to the side and take a pause for the good of everyone to assess rather than race to an uncertain future"

Emad Mostaque, *Founder and CEO of Stability AI*

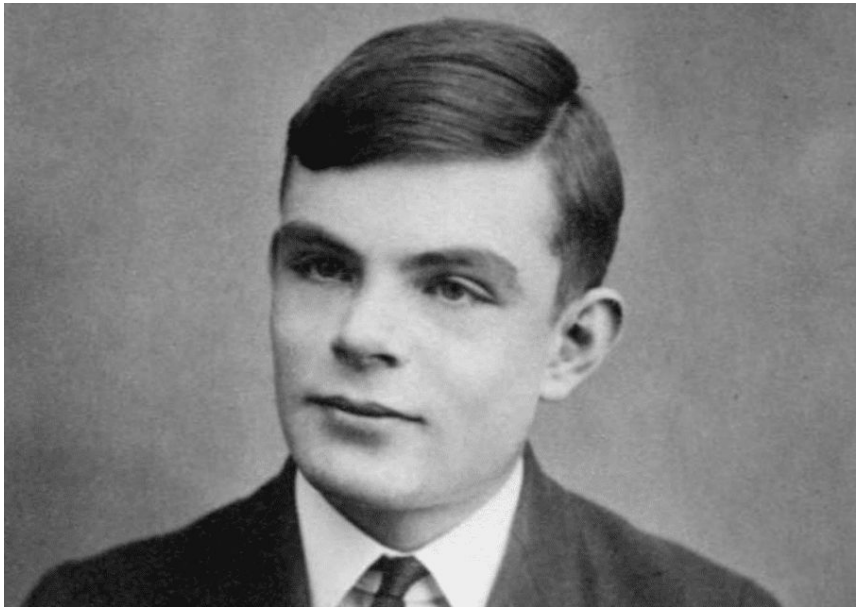


Policy recommendations:

1. Mandate robust third-party auditing and certification.
2. Regulate access to computational power.
3. Establish capable AI agencies at the national level.
4. Establish liability for AI-caused harms.
5. Introduce measures to prevent and track AI model leaks.
6. Expand technical AI safety research funding.
7. Develop standards for identifying and managing AI-generated content and recommendations.

“We can see only a short distance ahead, but we can see that much remains to be done.”

- Alan Turing (1950)



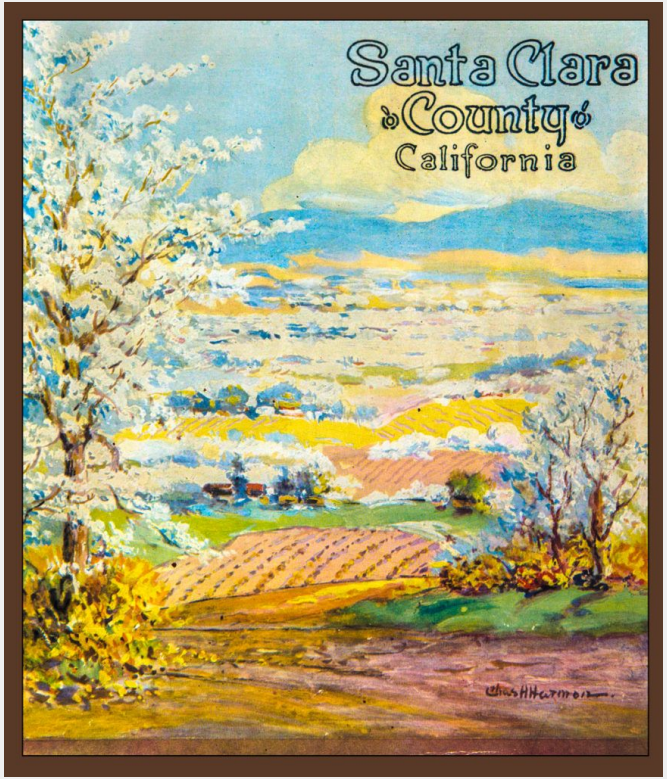
This transitions nicely into…

This week!

- What are the forces that created the modern Silicon Valley?
- Is AI/AGI creation, a self-fulfilling prophecy?
- How does ideology play a role in AI, tech development?
- How does finance?

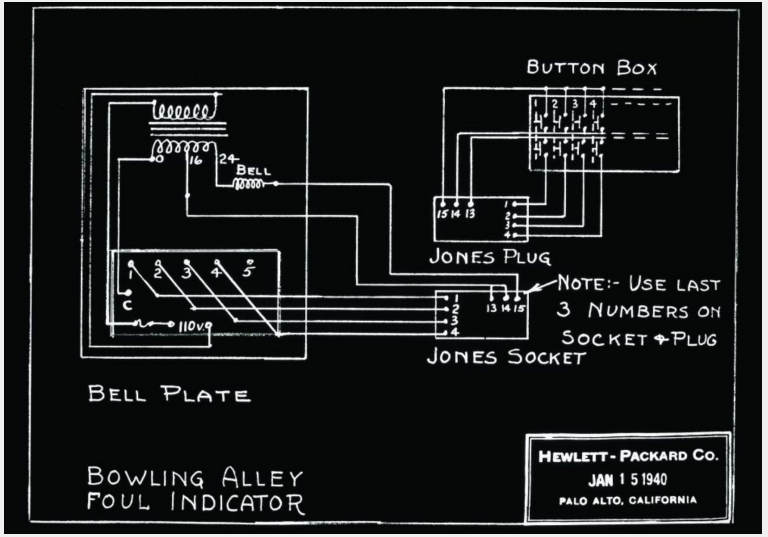
Silicon Valley before Silicon

- “Valley of Heart’s Delight”
- Mostly farmland with 40 canneries
- International prune export
- Delmonte, Sunsweet started here!
- Chinese, Japanese immigrants first, then Italian, East/Southern Europeans
- WWII: gradually replaced by Mexican immigrants (braceros)



HP 1939

- Hewlett & Packard built the HP 200A audio oscillator in their Palo Alto garage
- HP 200B custom made for Disney's Fantasia
- Stanford Prof, Fred Terman's mentorship ("Father of Silicon Valley")
- Pivot projects: automatic toilet flusher, an electric weight-loss machine and this device, an electric foul-line indicator they developed for a local bowling alley



Bowling Alley Foul Indicator

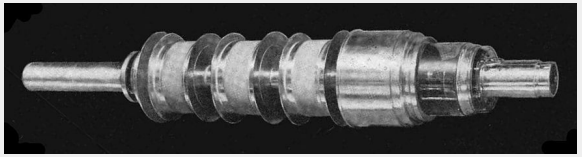


HP 200B audio oscillator

WWII comes to Silicon Valley

- Radar, electronics, communications
- Varian Associates (1948)
  - Made parts for radiation therapy, fuse for atomic bomb, etc.
  - Military contracts
- Stanford Research Park (1951):
  - partnership between Palo Alto and Stanford
  - Kodak, GE, Lockheed offices





Klystron (specialized vacuum tube)



Fred Terman “Father of Silicon Valley”

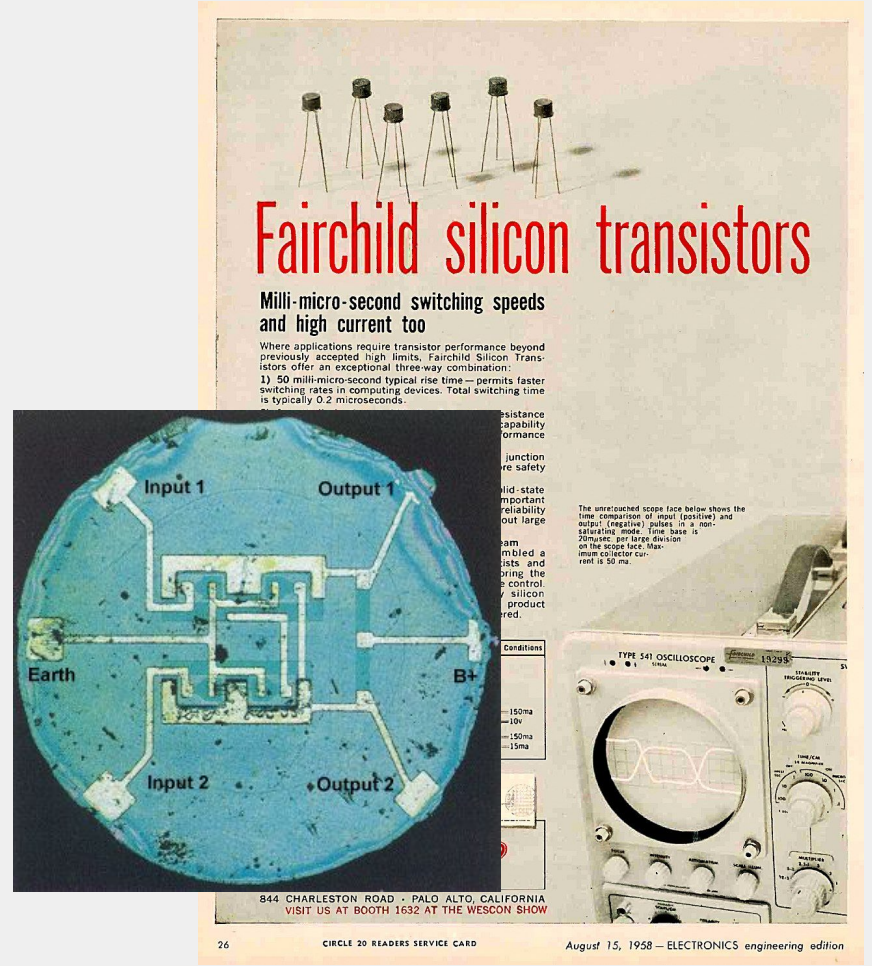
SRI → SRI International

- Established in the 1940s by Stanford for DARPA/gov subsidized projects.
- Broke off 1977 (due to Vietnam War protests)
- The first project was investigating whether guayule plant could be converted into sources of rubber. → rubber shortage/need for WWII
- 1960s: Shakey the Robot (robotics)
- Speech, networks, ARPAnet



Semiconductors

- In 1947 at Bell Labs, John Bardeen, Walter Brattain, and William Shockley invent transistors – can replace bulky vacuum tubes!
- Shockley Semiconductor 1956
- Fairchild Semiconductor 1957 by the “Traitorous Eight”
- IBM bought 2N697 for $150 apiece in order to build the computer for the B-70 bomber.
- Fairchild made the first silicon integrated circuit
- Fairchild spawns 60+ of spinoff semiconductor companies —e.g. Intel (1968, founded by Noyce and Moore)



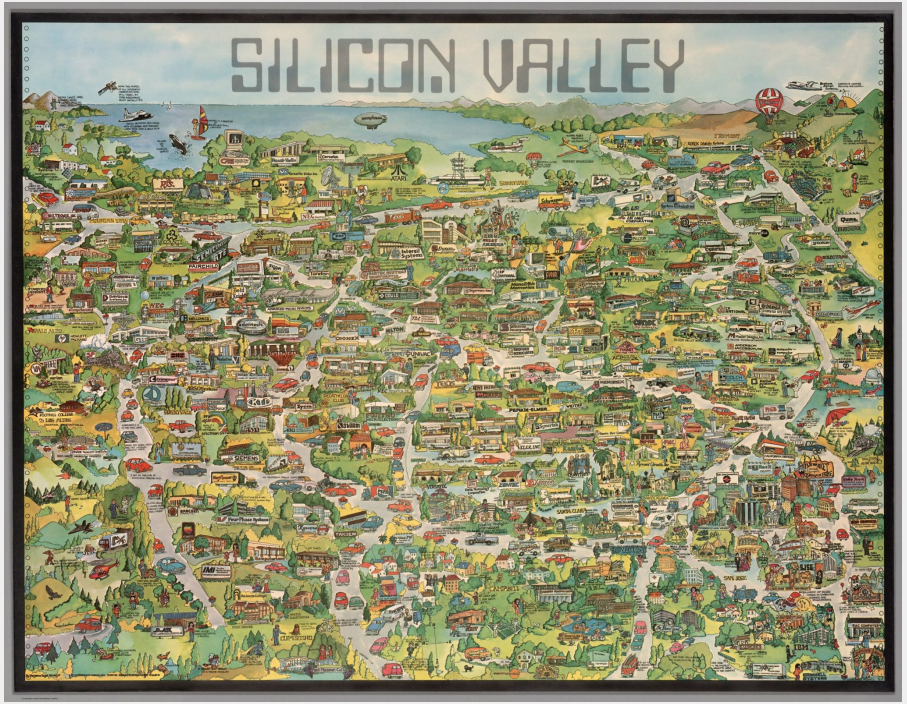
Silicon Valley VCs 1980s

- More money → more risk taking
- Heavy investment in computing, networking, internet
- Expert Systems in 1980s
- Capital Gains Tax cuts 1978, 49.5% to 28%
- Apple's billion dollar IPO in 1980 inspires VCs
- Previously VCs funded mostly semiconductor microchips → expand into biotech, software, lifestyle, AI
- ERISA "prudent man" rule pivot 1979 Pivot: 5–10% of pension funds into "alternative investments" like venture capital
- Sudden cashflow into VC: $4-5 billion in 1983



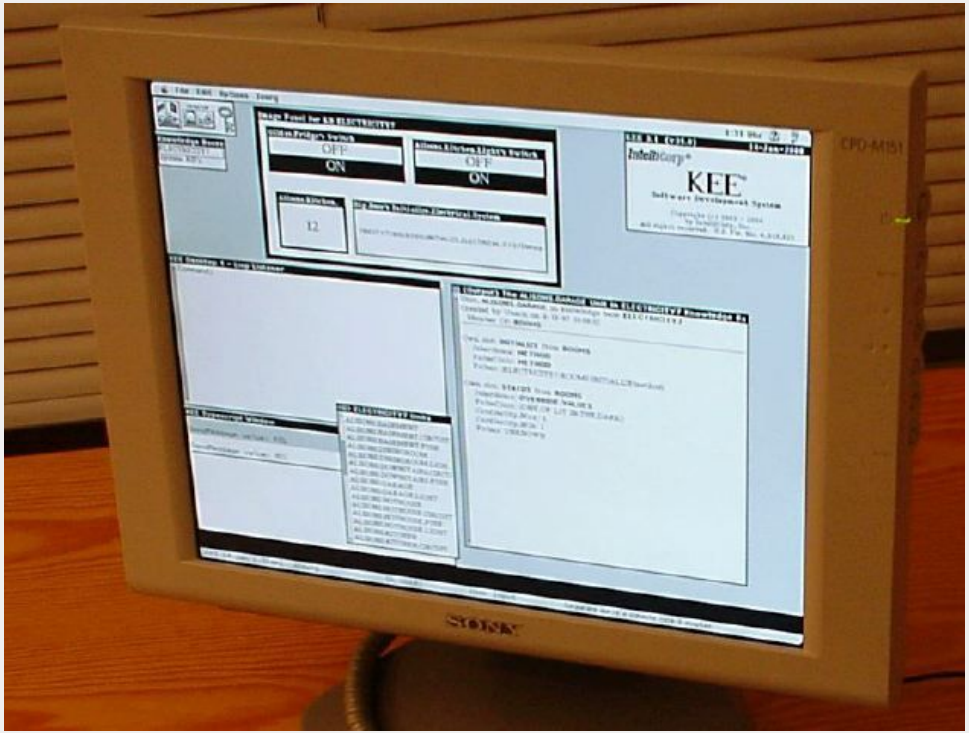
Expert Systems Start-ups 1980s

- Expert Systems was the first AI technology to arouse commercial interest
- Teknowledge, Inc., Intellicorp sprang up in the early 1980s to market these systems. Both these companies received millions from DARPA to bring AI into the market and military fields.

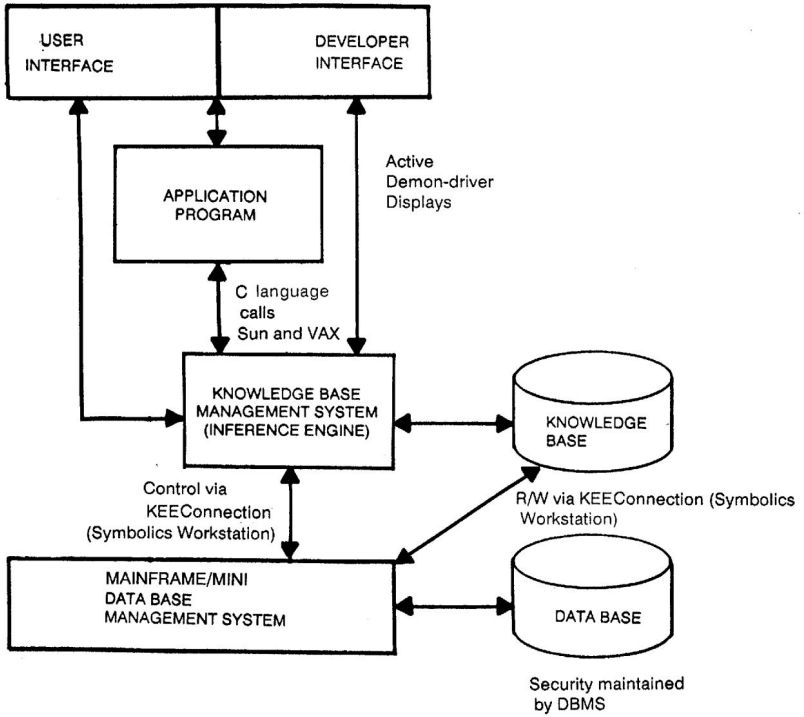


Knowledge Engineering Environment (KEE) 1983

- Intellicorp created and started selling Knowledge Engineering Environment (KEE) 1983: a high-end knowledge expert system development shell.
- KEE was to support knowledge acquisition and representation, reasoning, and user interface construction.
- Framework for multiple expert domains



Product Information



Software

- **KEE Version 3.0 for workstations**
- **KEE Version 3.0 for 80386-based IBM-compatible PCs**
- **KEEConnection**
- **SimKit**
- **Run-Time KEE**

KEE Versions 1.0 & 2.0 and Intelliscope are no longer sold.

Pricing

KEE Version 3.0 (workstations)	**$46,500** academic price **$9,250**
KEE Version 3.0 (386 PCs)	**$9,900**
KEEConnection (varies with hardware)	**$15,500** academic price **$5,000**
SimKit	**$21,000** academic price **$4,125**
Run-Time KEE (single-user workstation)	**$5,000**

Pricing varies, depending on number of licenses purchased, and includes 3-9 days training for 2 persons and one year's support. Support services include standard consultants, senior scientist consultants and/or apprenticeship training (4 weeks).

Hardware

all 80386-based IBM PC-compatible computers
IBM PC RT
Symbolics 36XX series
Xerox 1100 series
Sun 3 workstations
Apollo Domain workstations
TI Explorer I and II
DEC VAX series
MicroVAX II workstations

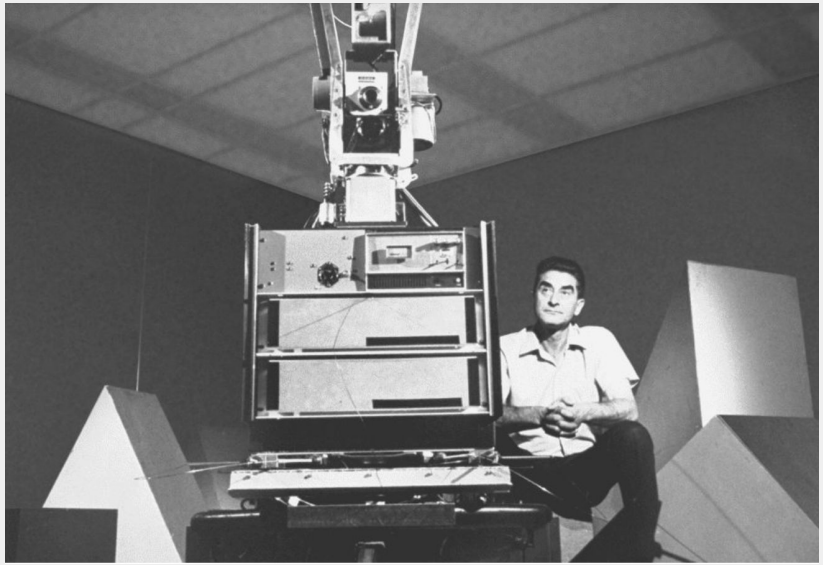
Silicon Valley VCs

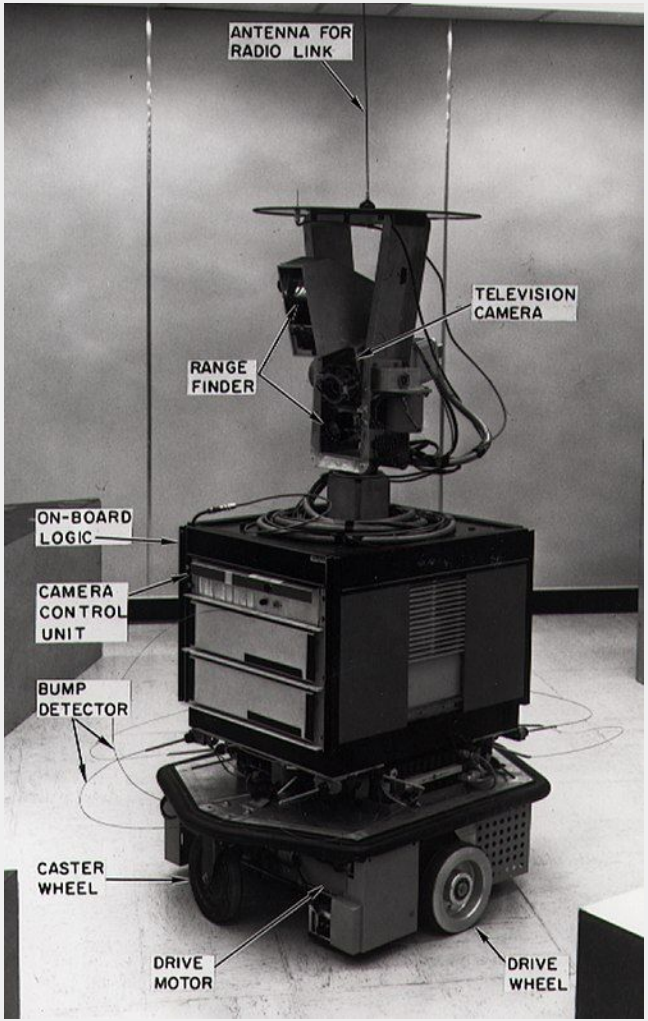
- Eugene Kleiner (Fairchild Semiconductor co-founder) and Tom Perkins found Kleiner Perkins 1972 - known for Genentech, Google, etc.
- Don Valentine, a semiconductor vet founds Sequoia Capital 1972 - Apple early investor, Cisco
- Bill Draper founds Draper Associates 1980s - Hotmail investor
- Dick Kramlich founds NEA 1977
- Accel Partners 1983

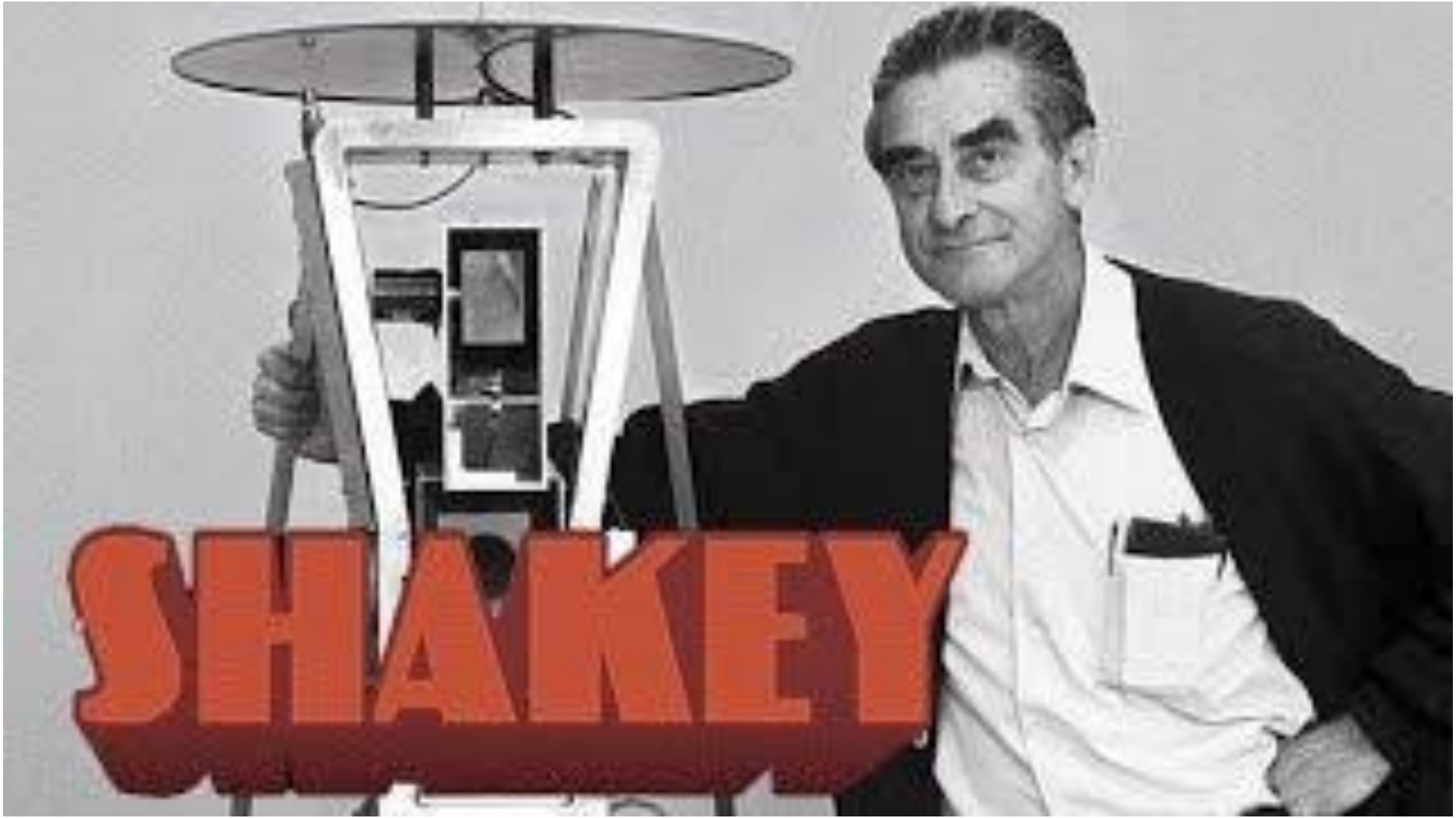


SRI Shakey

- AI center founded in 1966 - 1972
- First project: Shakey the robot, the “first electronic person”; STRIPS (the software program) written in LISP






SHAKEY

SRI Centibots 2003

- Centibots
- Research in swarm robotics and multi-agent systems
- DARPA sponsored $2.2m projects





Centibots 2003

SRI's SIRI (2000-2008)

- DARPA funds a project called CALO (Cognitive Assistant that Learns and Organizes) —> an assistant that can send emails helping users organize their tasks, etc.
- spin-off of (Personal Assistant that Learns) framework from DARPA, a defense assistant tool
- February 2010, Siri was released as an app for iOS – two months later Apple acquired it and integrated Siri software into its hardware devices.



SIRI 2010

Xerox PARC 1970

- R&D division of Xerox
- Laser printing, Ethernet, GUI, computer mouse



Dot Com Boom

- Search 90s: Netscape (1994), Yahoo (1995), and Google (1998) turned the Valley into a digital goldmine.
- Lifestyle 2000s: Facebook (2004), YouTube (2005), and the iPhone (2007)
- AI in the 2010s???



Deepmind 2010

- How to make AGI for the benefit of humanity
- “Could we create a machine that could work backwards to make sense of the universe? AGI would give you an insight into where we came from, and what is God.”
- Demis Hassabis, Shane Legg and Mustafa Suleyman cofounders
- Early investment from Peter Thiel, Elon Musk, Jaan Thallin
- Met investors at Singularity Summit
- Earliest projects on reinforcement learning with game screen pixels as input
- Ethics board



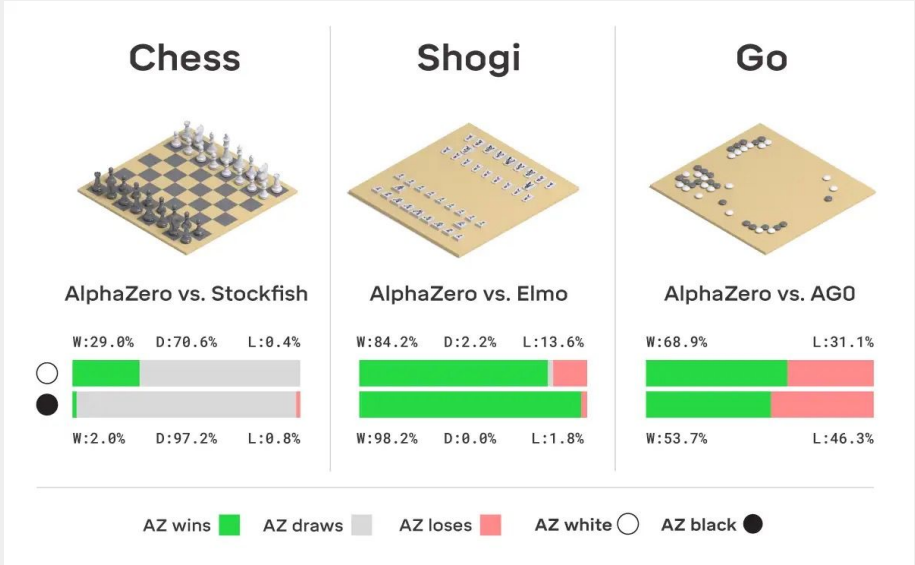


AlphaGo 2014, AlphaZero 2017



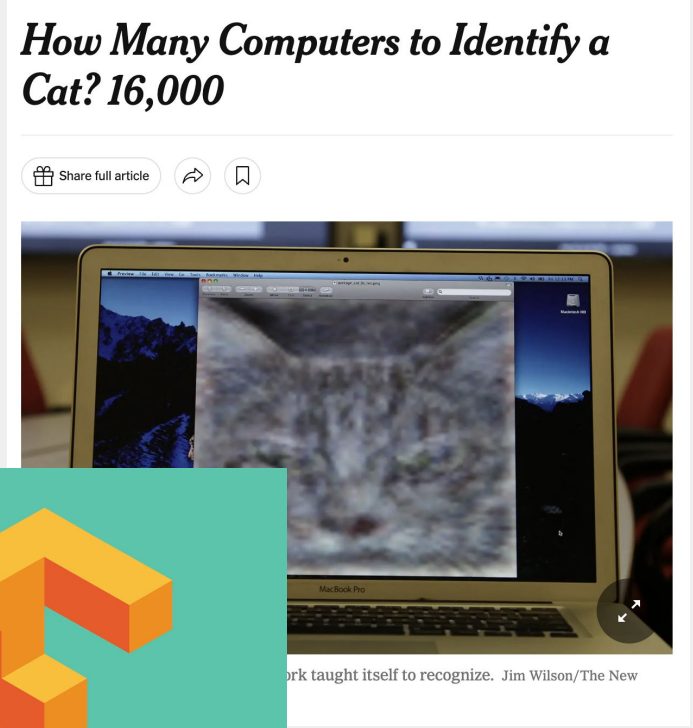
AlphaZero (2017) plays Chess, Shogi & Go
Self train

AlphaGo(2014) beat Lee Sedol 4:1 in a series of matches in 2016
30 million + moves learned



Google Brain 2011

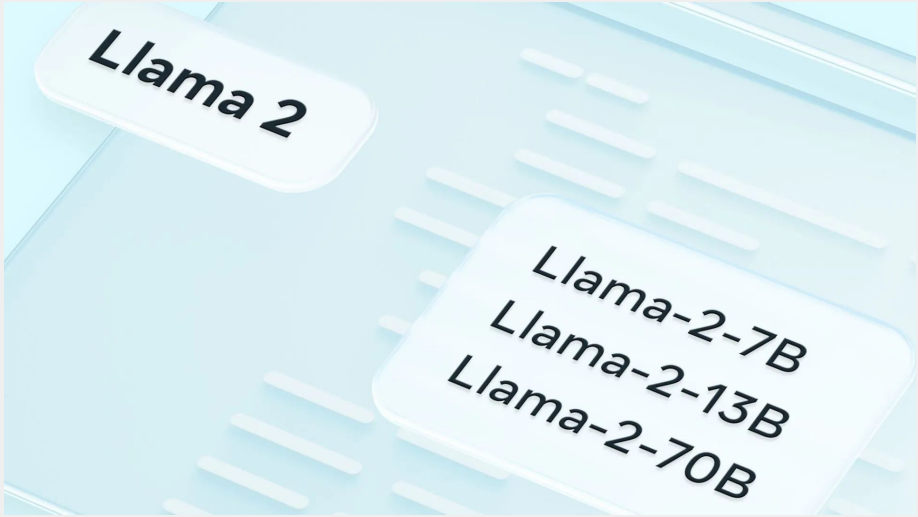
- Unsupervised “cat experiment” 2012: Stanford professor Andrew Ng. One of the group’s first projects was training a cluster of 16,000 processors in 1,0000 machines on neural nets to recognize a cat based in 10 million images from Youtube videos.
- Focus on Google Ads
- Customer AI Chips (TPUs 2016)
- 2015 Tensorflow 1.0





Facebook FAIR 2013

- Yann LeCunn heads FAIR and pushes open source
- PyTorch 2016
- RoBERTA 2019
- Llama series



PyTorch

OpenAI 2015

- Sam Altman et al.
- Some handful of Deepmind researchers
- Non-profit dedicated to AI safety
- Elon Musk wants to fund Deepmind rival, “Hitler of AI”
- Altman email to Musk on May 25, 2015: “someone other than Google” should build AGI first.



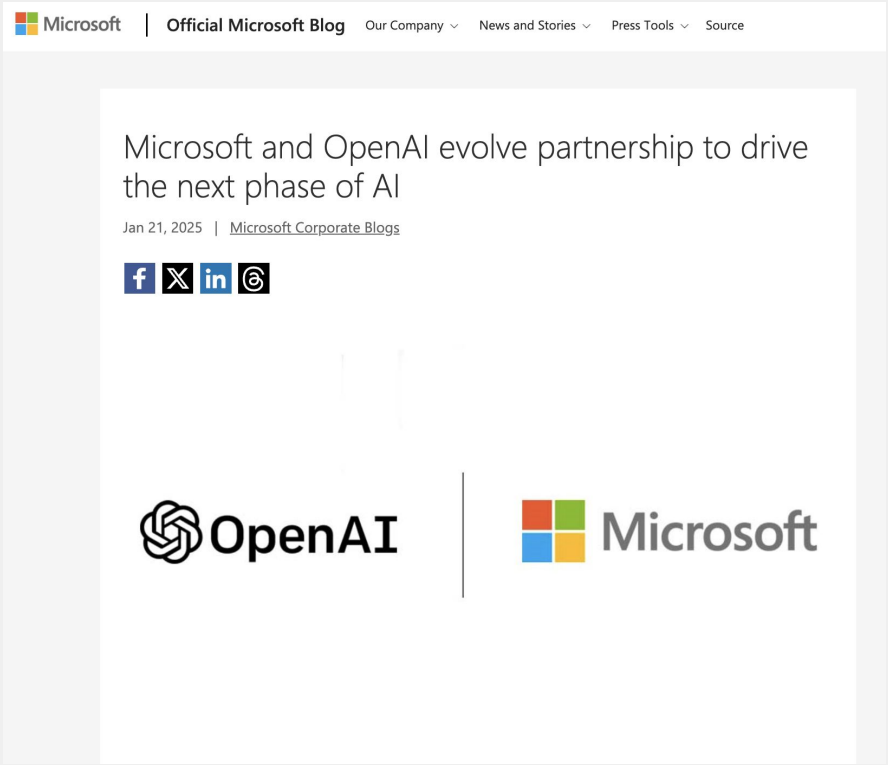


“Proxy Wars”

- Google vs. Microsoft masked as Deepmind vs. OpenAI?
- Google acquires Deepmind 2014 $650m
- Microsoft $1b investment 2019 in OpenAI (since then much more) - inspired by XiaoIce debacle
- OpenAI becomes “Capped profit”-able
- Deepmind aims for “prestige”, OpenAI goes more “product”

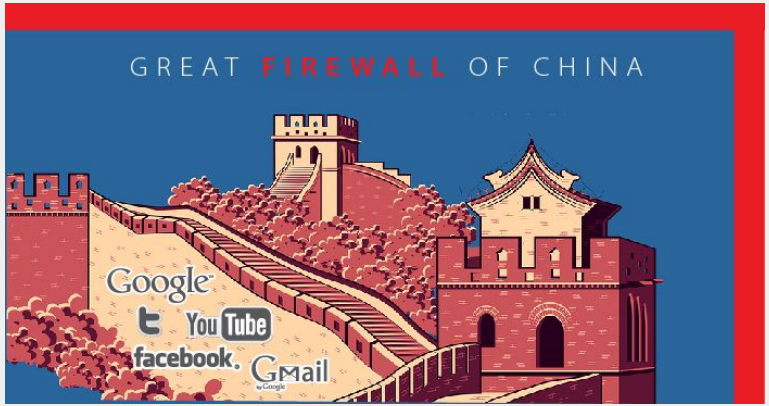


XiaoIce



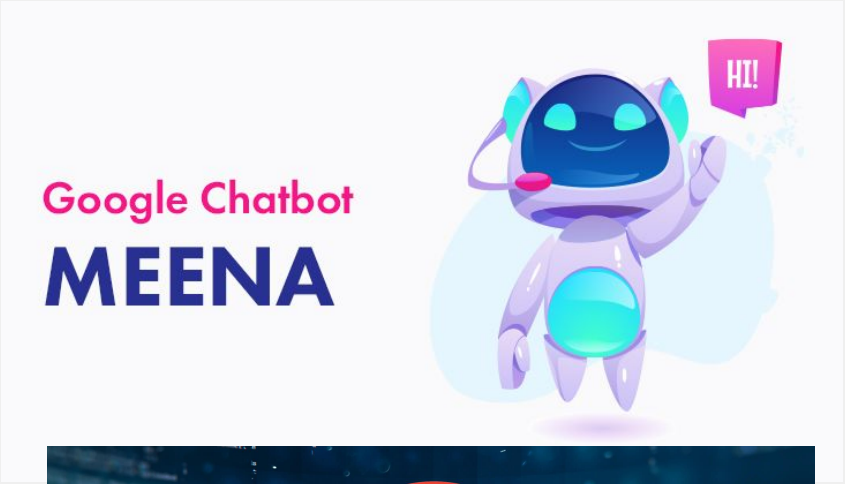
Google locked out of China 2010

- Google focused on re-entering China
- In 2010 it had exited the country after accusing Beijing of hacking its intellectual property and the Gmail accounts of Chinese human rights activists.
- The Chinese government demanded that Google censor searches about Tiananmen Square and other controversial topics for the Chinese Communist Party
- China blocks access to Facebook and Twitter, setting up what became known as the Great Firewall
- Project Dragonfly (Google search for China)
- Deepmind Alpha Go Master beats Chinese play Ke Jie 3 matches but not aired in China



Google’s Transformers

- Google Brain’s Vaswani et al released transformer model based on attention mechanism only
- But Google too cautious any new experiments will hurt their $100 billion ad product
- Not really applied anywhere except BERT embeddings and machine translation





Anthropic 2021

- Delaware Public Benefit Corporation
- Dario Amodei & Daniela Amodei from AI Safety team OpenAI
- 7 former OpenAI employees
- Funds from Google, Amazon
- FTX founders support out of effective altruism



ChatGPT 2022

- Just the decoder part of Transformers?
- GPT, GPT2, GPT3, DALL-E 2
  - Gradually less transparent!
- After two months online ChatGPT had attracted thirty million registered users, (one of the fastest-growing online services in history)
- By early 2024, around one hundred million people.



Aug 29, 2024 - Technology

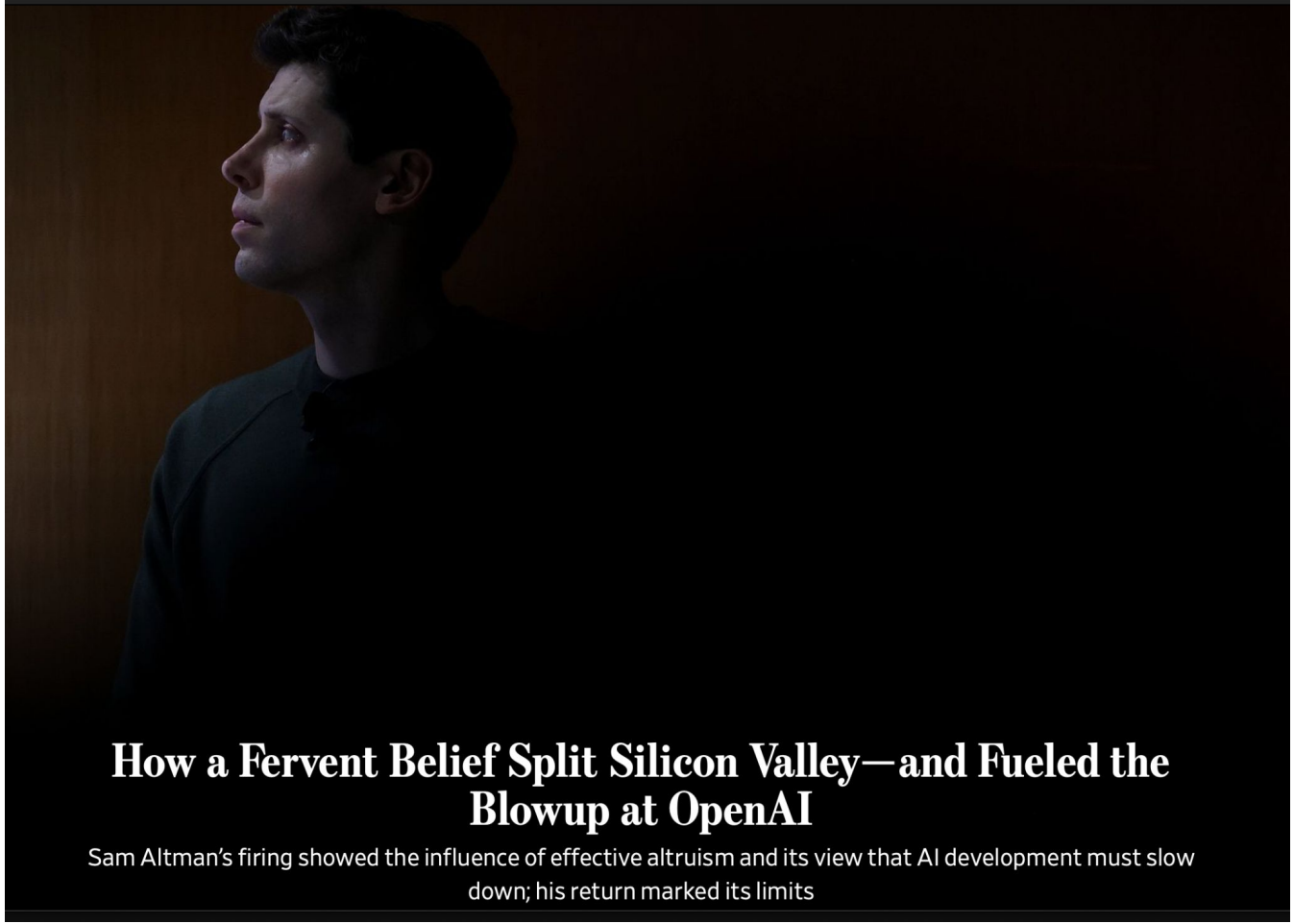
OpenAI says ChatGPT usage has doubled since last year

Ina Fried

OpenAI said on Thursday that ChatGPT now has more than 200 million weekly active users — twice as many as it had last November.

THE WALL STREET JOURNAL.

SIGN IN



How a Fervent Belief Split Silicon Valley—and Fueled the Blowup at OpenAI

Sam Altman's firing showed the influence of effective altruism and its view that AI development must slow down; his return marked its limits

Google AI
2017

DeepMind
2014

Google
Brain
2011

Google
DeepMind
2023

